No-Extra Components Density Map Cropping Guided Object Detection in Aerial Images

Zhe Guo, Guoling Bi, Hengyi Lv, Yang Feng, Yisa Zhang, and Ming Sun

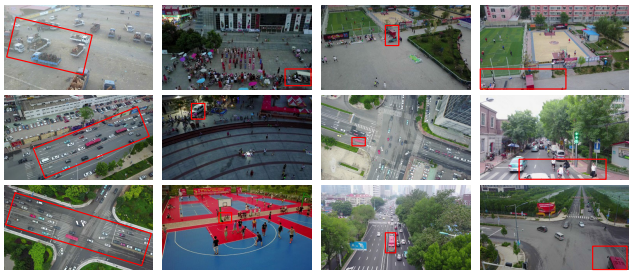
***Abstract*— Aerial images usually contain a large number of truncated and small objects, which poses a significant challenge for object detection. Existing methods have introduced additional learnable components in the pipeline and adopted multistage training approaches, but they have not solved the problem of achieving end-to-end detection. To address this issue, we propose a novel no-extra components density map cropping (NE-CDMNet) method to utilize the spatial and contextual information between objects to improve detection performance. Furthermore, we introduce a new query selection (QS) scheme that utilizes confidence scores to select the top-K features from the encoder, helping the model better leverage the position information for extracting more comprehensive content features. Finally, we incorporate the local–global fusion (LGF) algorithm to combine the detection results from the original image and the density-cropped image. We conducted extensive experiments on two widely used public aerial datasets. Results reveal that the proposed method achieves the best performance compared with other state-of-the-art methods, whose 35.3% AP on the VisDrone-DET2019 dataset and 79.7% mAP on object detection in optical remote sensing image (DIOR) dataset, demonstrate the effectiveness of our method.**

***Index Terms*— Aerial images, density map cropping (DMC), fusion detection, no-extra components, object detection, query selection (QS).**

I. Introduction

OBJECT detection, as a fundamental task in Earth vision, is crucial for Earth detection [1], monitoring [2], [3], inspection [4], [5], [6]. In recent years, with the development of deep neural networks (DNNs), methods such as FRCNN [7], RetinaNet [8], and Mask region-based convolutional neural network (R-CNN) [9] have become standard pipelines for general object detection. However, these methods, although achieving good performance on natural image datasets, such as MS-COCO [10] and Pascal VOC [11], often fail to deliver satisfactory results on aerial image datasets.

Given that aerial images are typically captured from a bird's-eye view using aircraft, such as drones, planes, or satellites, they exhibit substantial scale differences compared with natural images like those found in ImageNet [12]. These unique characteristics pose significant challenges for object detection in aerial images: 1) due to the aerial perspective, objects in bird's-eye view images are frequently oriented in various directions and distributed rotationally; 2) aerial images commonly contain numerous small objects that are sparsely distributed within densely populated areas of objects; 3) in many instances, there is a significant imbalance in the distribution of objects across different categories; and 4) aerial images frequently encounter challenges such as occlusions between objects and truncation at object boundaries, as shown in Fig. 1.



(a) (b) (c) (d)

Fig. 1. Challenges in object detection in aerial images (highlighted in red), example images taken from VisDrone2019 dataset. (a) Arbitrary orientation of objects is caused by the overhead view. (b) Small objects are sparsely distributed within densely populated areas of objects. (c) Number of object classes is unbalanced. (d) There are occlusions between objects and truncation of object boundaries.

In the early stages of object detection in aerial images, researchers primarily employed general object detection architectures and emphasized enhancing the object detection performance [see Fig. 2(a)]. Ding et al. [13] introduce an region of interest (RoI) Transformer, the core idea of which is to apply spatial transformation on the RoIs and learn the transformation parameters under the supervision of oriented bounding box (OBB) annotations. Yang et al. [14] utilize an adversarial learning framework to explicitly separate task-specific and nuisance features in a supervised manner, facilitating the learning of task-specific, domain-invariant features. Wu et al. [15] design a sampling fusion network to fuse multilayer features with effective anchor sampling to improve sensitivity to small objects. The improved object detection method achieves reasonable results on popular datasets [1], [2], [3], [4] they are far from satisfactory for practical applications.

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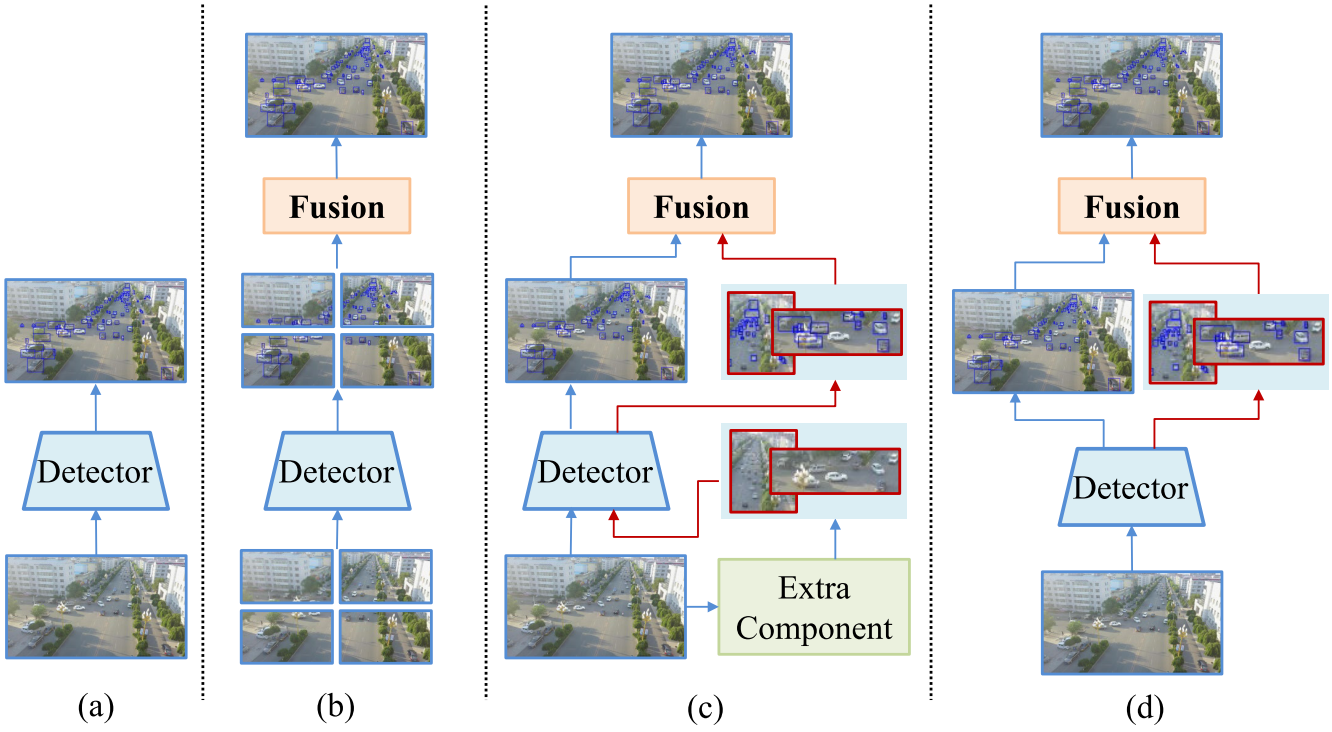


Fig. 2. Illustration of aerial image object detection methods at different stages. (a) General object detection architectures. (b) Original aerial image into uniform crops or pieces before applying object detection. (c) Selectively crop the original image through an additional learnable module before applying object detection. (d) Our algorithm, which no extra components to selectively process the original image.

To tackle the issue of scale variation, another promising approach is to divide the original aerial image into smaller crops or pieces before applying object detection. This can be achieved through techniques such as uniform cropping [16] or random cropping [see Fig. 2(b)]. In many instances, these straightforward cropping strategies prove effective in enhancing the detection accuracy of small objects. This is primarily due to the fact that the crops obtained from cropping are subsequently resized to match the original image size, thereby achieving a higher resolution for training. However, these methods lack the ability to utilize semantic information to guide the cropping process, which limits their effectiveness in effectively separating the foreground and background during cropping. In addition, larger objects in the aerial images often get truncated and divided into two or more separate crops.

Building upon the concept of cropping, the method used to determine appropriate crop regions is crucial for achieving accurate object detection in aerial images. It is evident that selective-based cropping strategies yield superior results compared with uniform or random strategies. To extract these density crops, existing methods utilize an additional learnable module to process the original image [see Fig. 2(c)]. ClusDet [17], DMNet [18], and coarse-grained density map network (CDMNet) [19] employ a multistage training approach by introducing additional learnable components into the pipeline. Initially, the high-resolution image is divided into multiple subregions, and object detection is performed on these subregions. The detection results from the subregions are then fused with the detection outputs obtained from the original image, resulting in the final detection result.

In this article, we attempt to focus on foreground objects by proposing an end-to-end selective crop detection method that utilizes density map intensity in a standard detector to determine the dense object presence regions. Provides efficient and accurate cutting without requiring the involvement of additional components [see Fig. 2(d)].

We summarize our main contributions as follows.

1) We are the first to propose a no-extra components density map cropping (DMC) method to guide image cropping. First, it maps objects to their corresponding positions on the density map. Then, it utilizes a normalized Gaussian kernel to represent the distribution of objects on the density map accurately. Finally, the density map is used to generate cropped images using a binary method.
2) To address the difficulty of optimizing object queries, we design a new query selection (QS) scheme to help the model better utilize the position information to obtain more comprehensive content features from the encoder.
3) We propose a local–global fusion (LGF) algorithm that aims to combine the detection results from the original image and the density cropping image.

The rest of this article is organized as follows. Section II discusses the related work for object detection. Section III presents the methodology in detail. Section IV provides experimental results on two datasets and extensive ablation studies. Finally, Section V concludes this article.

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II. Relate Works

A. General Object Detection

General object detection targets primarily on natural scenes. According to the detection pipeline, existing detectors can roughly be categorized into two types: one-stage networks and two-stage networks. One-stage networks are designed to enhance detection speed, such as single shot multibox detector (SSD) [20], you only look once (YOLO) series [21], [22], [23], and RetinaNet [8], which directly predict the location of objects without an explicit region proposal stage. In contrast, the two-stage network added a region proposal stage to refine the target of the candidate region to obtain more accurate object categories and locations, such as Fast R-CNN [24], FRCNN [7], and Cascade R-CNN [25].

While existing object detection methods exhibit superior performance in natural scenarios,(e.g., 500 × 400 images in Pascal VOC [11] and 600 × 400 images in MS-COCO [10]), they often fall short in achieving satisfactory results when applied directly to aerial scenarios(e.g., 1080 × 540 images in UAVDT [2] and even higher resolution aerial datasets).

B. Object Detection in Aerial Images

Compared with object detection in natural images, object detection in aerial images faces greater challenges due to the existence of a large number of small objects. First, objects tend to exhibit smaller scales relative to high-resolution aerial imagery. Second, the distribution of small objects in aerial images is often sparse, uneven, and concentrated in specific areas. These unique characteristics of aerial images require specialized methods to overcome challenges and obtain accurate object detection results.

Considering that small objects are predominantly concentrated in the shallow layers of the feature pyramid, their feature representation may lack sufficient semantic information. To address this, an effective approach is to leverage contextual information to enhance the feature representation of small objects in these shallow layers. Zhang et al. [26] used a search algorithm based on the Bayesian optimization that can find more accurate object bounding boxes without significantly increasing the number of bounding box proposals. Wang et al. [27] used nonlocal operations to compute the response at a position as a weighted sum of features at all positions to capture a generic family of building blocks for long-range dependencies. Hu et al. [28] presented a context-aware RoI pooling to maintain the contextual information and original structure of small-scale objects. Li et al. [29] proposed a solo-to-collaborative dual attention network to enhance shallow small object feature representation. Zhang et al. [30] designed a composite semantic feature fusion architecture that can set coupled or uncoupled semantic relationships for each pixel, which is beneficial to accurate object center point prediction.

Contextual information are crucial for small object detection. However, as the depth of the network increases, the resolution of the feature map decreases, making it difficult to accurately capture the location information of the target. Some literature improves the detection accuracy of small objects by designing multiscale feature pyramids. The most classic method, FPN [31] built a top–down architecture with lateral connections for building high-level semantic feature maps at all scales. Liu et al. [32] enhanced the entire feature hierarchy with accurate localization signals in lower layers by bottom–up path augmentation, which shortens the information path between lower layers and topmost features. Ghiasi et al. [33] designed a feature pyramid via neural architecture search, consisting of a combination of top–down and bottom–up connections to fuse features across scales. Tan et al. [34] proposed a weighted bidirectional feature pyramid network to allow multiscale feature fusion to be performed more easily and quickly. Peng et al. [35] proposed a shortcut feature pyramid network, by adding some shortcut connection paths based on residual learning to the traditional feature pyramid network enables the spatial location features at the bottom of the convolutional network and the semantic features at the upper level to complement each other, making up for the information lost after multiple operations.

Recently, Yang et al. [17] proposed a clustered detection (ClusDet) network to cluster subregions of original images for fine detection. Li et al. [18] proposed a density-map guided object detection network (DMNet) that exploits pixel intensity changes to determine whether to crop an object in a given region of the image. Deng et al. [36] proposed a global–local self-adaptive network that utilizes local fine detectors to adaptively refine the object bounding boxes detected by global coarse detectors to achieve higher resolution detection. Duan et al. [19] proposed a CDMNet that crops the input image into multiple subregions through a cluster region generation algorithm guided by density maps to solve the problems of small targets and uneven distribution in aerial image detection. Most recently, Meethal et al. [37] proposed an efficient cascade amplification detector. It first adds an additional class of "density crops" to the detector, then performs inference on the original image and the up-scaled high-quality crops, and finally fuses the detection results. Although existing selective cropping methods have shown promising results, they often rely on multistage methods to guide image cropping. To this end, we focus on designing a method without additional components to achieve guided aerial image target detection.

III. Method

In this section, we detail the no-extra components density map cropping guided method. We first describe the framework overview of the method and then introduce the implementation of our proposed DMC, QS scheme, and LGF algorithm.

A. Overview

As shown in Fig. 3, The proposed method consists of three components: DMC module, QS scheme, and LGF algorithm. In detail, first, we generate density maps of aerial images and process them using soft masks and binary masks. Next, we feed the cropped image and the original image into the detector. To further improve the performance of the model, we adopt the QS scheme to help the model better utilize the position information to obtain more comprehensive content

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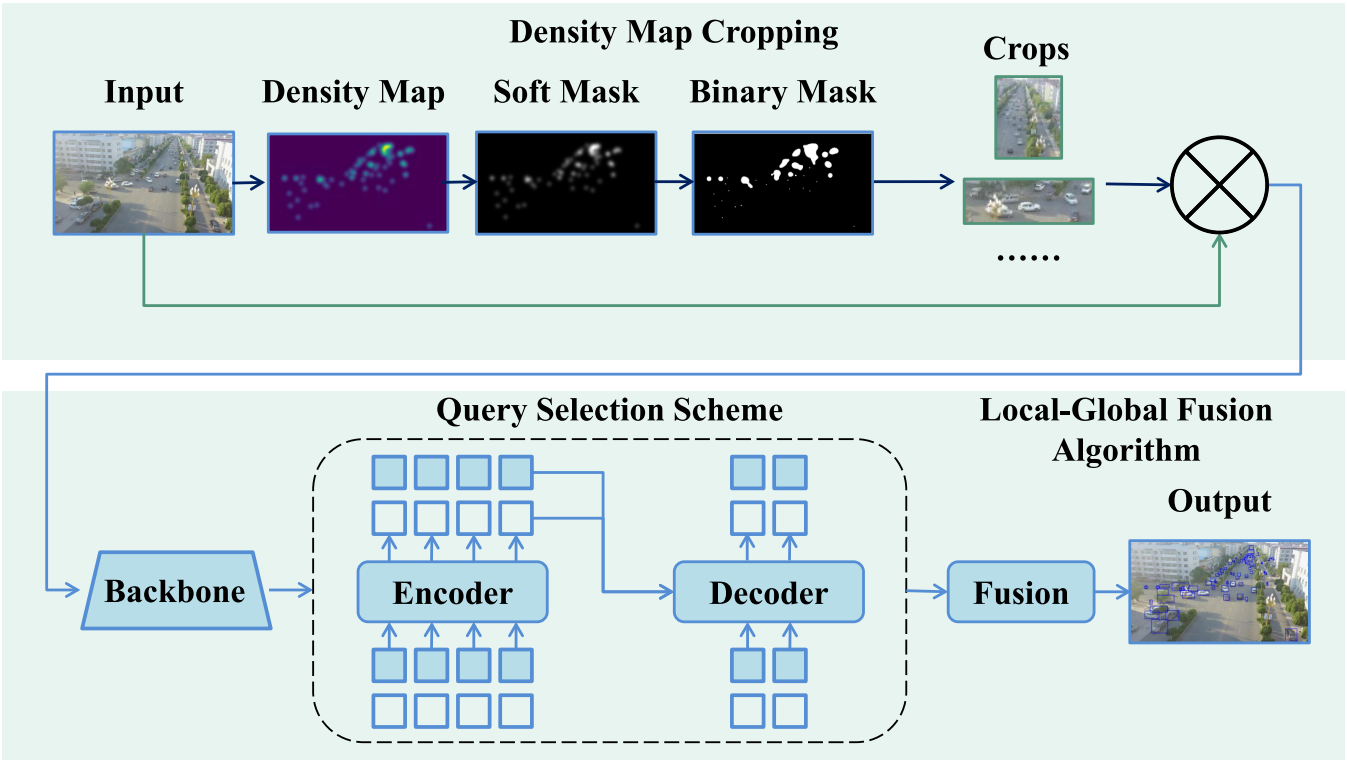


Fig. 3. Framework of our method. We first input the aerial images into DMC to generate the proposed regions for cropping. The generated image crops and the original image are fed into the same object detector for object detection. Next, QS is used during training to obtain more comprehensive content features from the encoder. Finally, the fusion prediction result is output.

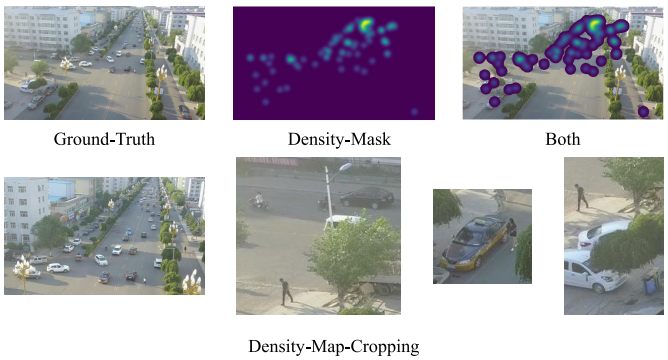


Fig. 4. Visualization of DMC.

features from the encoder. Finally, the detection results are fused using an LGF algorithm.

B. Density Map Cropping

Density map generation is the core of DMC. It was often used in crowd-counting tasks, helping to obtain more accurate and comprehensive information. MCNN [38] proposed to use multicolumn CNN to learn density maps for crowd-counting tasks. This is because, due to different head sizes in different images, a single column with a fixed receptive field may not capture enough features. To address this challenge, MCNN [38] introduced three columns, each with different receptive field sizes, to enhance the feature extraction capabilities of the network. By leveraging this multiscale feature representation, MCNN [38] was better able to handle scenes with different crowd density levels, resulting in improved crowd-counting performance compared with single-column method.

In order to generate a ground-truth density map, we follow a similar method to the one used in MCNN [38]. First, we map the objects to their corresponding positions on the density maps. Assuming that the coordinate of the object center point is (x_c, y_c) in the input images, the corresponding position on the density maps are calculated as shown in the following equation:

$$x_i, y_i = \frac{x_c}{\alpha}, \frac{y_c}{\alpha} \tag{1}$$

where α is the down-sampling factor of density maps relative to input images. We use fixed kernels and geometrically adaptive kernels to process input images, blurring each object annotation with a Gaussian kernel normalized to 1 to generate ground-truth density maps, as defined in the following equations:

$$D(x, y) = \sum_{i=1}^{N} G_{\sigma_i}(x - x_i, y - y_i) \tag{2}$$

$$\sigma_i = (\beta H_i, \beta W_i) = \frac{1}{2}\sqrt{H_i^2 + W_i^2} \tag{3}$$

where N is the object number of images. G_{σ_i} is the Gaussian kernel, which convolve with $(x-x_i, y-y_i)$ to generate ground-truth density mask. H_i and W_i are the height and width of the object i on density maps. β is configured to be 0.15 in our experiment. We use a soft mask to adjust the density mask

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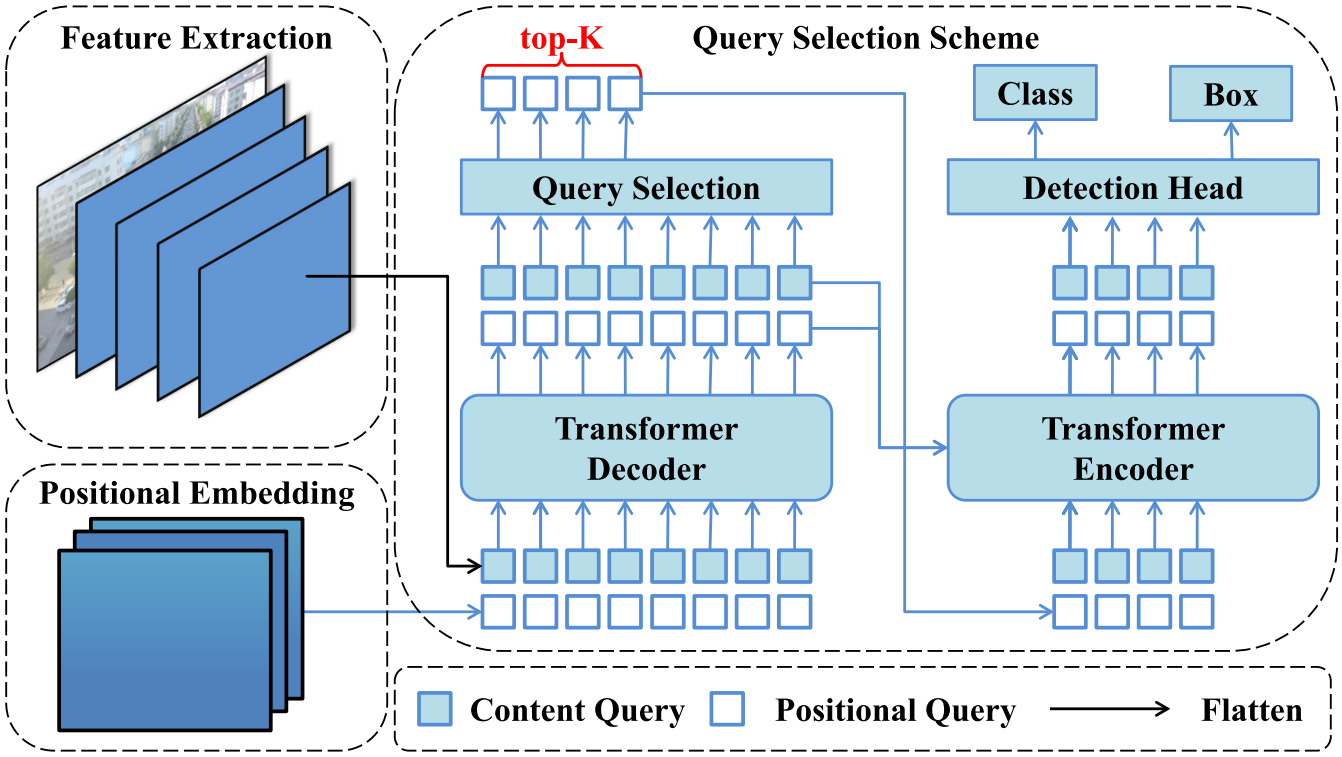


Fig. 5. Framework of QS scheme.

area and better extract the foreground objects. The soft mask can be generated using the following equation:

$$S(i, j) = \text{Normlization}(D(i, j)). \tag{4}$$

We introduce binary mask on soft mask by traversing H and W. If the pixel intensity is greater than the threshold, the pixels in the window will all have a value of 1; otherwise, they will be 0. This produces a binary mask with 0 and 1 values. The detailed implementation is illustrated in Algorithm 1.

Finally, we generate image crops based on binary masks. All pixels corresponding to a binary mask value of "1" are selected, and eight neighbor-connected pixels are merged into a larger candidate area. We then use the bounding rectangle of the candidate area to crop the original image, as shown in Fig. 4.

C. QS Scheme

End-to-end object detectors are known for their streamlined pipeline. In 2020, Carion et al. [39] first proposed the Transformer-based end-to-end detector DETR, which achieved good performance and effectively eliminated many hand-crafted components, including nonmaximum suppression and anchor generation. However, DETR [39] still faced slow convergence, high computational cost, and difficult query optimization issues.

To address the difficulty of optimizing object queries, we design a new QS scheme. It's structure is shown in Fig. 5. This scheme is different from DETR [39]. We use the confidence score to select the top-K features from the encoder to initialize the positional query. In addition, since the selected features are preliminary content features without further refinement, they may be ambiguous and misleading to the decoder. In order to better initialize the query, we only enhance the position query with the top-K selected features, keeping the content query as learnable as before. Its attention can be expressed as follows:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{5}$$

$$Q = Q_f + Q_p$$
$$K = K_f + K_p, \quad V = V_f \tag{6}$$

Algorithm 1 DMC

Input: Aerial image $X \in \mathbb{R}^{C\times H\times W}$
Output: Cropping image $Y \in \mathbb{R}^{C\times H\times W}$

```
Calculate density mask D(i, j) by Eq. 2
Calculate soft mask S(i, j) by Eq. 4
for i to H do
    for j to W do
        if S(i, j) ≥ thershold then
            B(i, j) = 1
        else
            B(i, j) = 0
        end if
    end for
end for
Match the binary mask area with the original image to obtain a new image and label
Y = convert(B(i, j), X)
return Y
```

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where d_k is the channel dimension, f means the feature, and p means the position embedding. The Q, K, and V are the query, key, and value respectively.

There are two kinds of attention in our method, one in the encoder and the other in both the encoder and the decoder. In the encoder, K_f and V_f are the same as Q_f while K_p is the same as Q_p. $Q_f \in \mathbb{R}^{N_q \times C}$ and $K_f \in \mathbb{R}^{N_q \times C}$ are generated by the linear transformation of the selected features. $Q_p \in \mathbb{R}^{N_q \times C}$ uses a set of learned embedding as follows:

$$Q_p = \text{Embedding}(N_q, C). \tag{7}$$

Another attention in the decoder, $Q_f \in \mathbb{R}^{N_q \times C}$ is a set of learnable content query while $K_f \in \mathbb{R}^{N_q \times C}$ and $V_f \in \mathbb{R}^{N_q \times C}$ are the output feature of the encoder. $Q_p \in \mathbb{R}^{N_q \times C}$ is the top-K features from the encoder using the confidence score through QS. $K_p \in \mathbb{R}^{HW \times C}$ is the position embedding of K_f. It is generated by the sine–cosine position encoding function g_{sin} based on the corresponding key feature position $\text{Pos}_k \in \mathbb{R}^{HW \times 2}$

$$K_p = g_{\text{sin}}(\text{Pos}_k) \tag{8}$$

where H, W, and C are the height, width, and channel of the feature, respectively, and N_q is the predefined number of the query.

D. LGFAlgorithm

After obtaining the detection results from the detector, the next step is to fuse the results of the original image and the density cropped image. Specifically, we first perform the separate detection on the original image and the density-cropped image. Then, we collect the predicted bounding boxes and classes from the density-cropped image detection and add them back to the original image's detection to fuse them. Finally, we apply the LGF algorithm to all bounding boxes and calculate the final result. For fusion detection, the threshold of the LGF algorithm was set to 0.55 in all datasets. It is worth noting that any existing modern detector can serve as our detector. The implementation of the LGF algorithm is summarized in Algorithm 2.

We use the LGF algorithm to fuse our prediction results. Then, we calculate the box coordinates and confidence scores in F. Formally, it can be expressed as follows:

$$\mathbf{C} = \frac{\sum_{i=1}^{\mathbf{T}} \mathbf{C}_i}{T} \tag{9}$$

$$\mathbf{X}_{1,2} = \frac{\sum_{i=1}^{\mathbf{T}} \mathbf{C}_i * \mathbf{X}_{1,2i}}{\sum_{i=1}^{\mathbf{T}} \mathbf{C}_i} \tag{10}$$

$$\mathbf{Y}_{1,2} = \frac{\sum_{i=1}^{\mathbf{T}} \mathbf{C}_i * \mathbf{Y}_{1,2i}}{\sum_{i=1}^{\mathbf{T}} \mathbf{C}_i} \tag{11}$$

$$\mathbf{C} = \mathbf{C} * \frac{\min(T, N)}{N} \tag{12}$$

where T is the number of bounding boxes, X_1, X_2, Y_1, and Y_2 are the coordinates of the bounding box, and C is the confidence of the fusion box.

Algorithm 2 LGF

Input: B a list of the predicted bounding boxes
Output: F a list of fuse results

```
Create a empty list L for boxes clusters
for i in B do
    for j in F do
        Calculate IOU(B_i, F_j)
        if IOU ≥ thershold then
            L_j.append(B_i)
            F_j = Fusion(L_j)
        else
            j = j + 1
            F_j = L_j = B_i
        end if
    end for
end for
return F
```

IV. Experiments

In this section, we first describe two widely used public datasets (VisDrone-DET2019 [3] dataset and DIOR dataset [4]) in aerial image object detection, implementation details, and evaluation metrics. Then, all experiments are analyzed quantitatively and qualitatively to validate the effectiveness of our no-extra components density map cropping (NE-CDMNet). All experiments were conducted on a GeForce RTX 3090 GPU with 24 GB of memory.

A. Datasets

1) VisDrone-DET2019 [3] consists of 8599 images captured by drone platforms in diverse locations and at varying altitudes. The dataset is partitioned into training, validation, and testing subsets (6471 for training, 548 for validation, and 1580 for testing), with the images collected from different locations but similar environments. The dataset includes more than 540 000 annotated bounding boxes covering ten predefined object categories (pedestrian, people, bicycle, car, van, truck, tricycle, awning tricycle, bus, and motor).
2) DIOR [4] is a large-scale object detection dataset for optical remote sensing images, containing 23 463 images and 192 472 object instances, covering 20 object classes, including airplane (APL), airport (APO), baseball field (BF), basketball court (BC), bridge (BR), chimney (CH), dam (DAM), expressway service area (ESA), expressway toll station (ETS), golf course (GC), ground track field (GTF), harbor (HA), overpass (OP), ship (SH), stadium (STA), storage tank (STO), tennis court (TC), train station (TS), vehicle (VE), and windmill (WM). It exhibits extensive object scale variation, not only in spatial resolution but also in intra- and inter-class size variation.

B. Implementation Details

We implement our network based on PyTorch [40] and MMDetection tool-box [41]. Stochastic gradient descent

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TABLE I
RESULTS OF OBJECT DETECTION ON THE VISDRONE-DET2019 VAL SET

Method	#epochs	#img	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	Params	GFLOPS	FPS
FRCNN-FPN-EIP [7]	12	3288	25.7	48.4	24.1	-	-	-	-	-	3.03
ClusDet-R50 [17]	18	2716	26.7	50.6	24.7	17.6	38.9	51.4	42M	137	3.66
ClusDet-R101 [17]	18	2716	26.7	50.4	25.2	17.2	39.3	54.9	60M	186	2.73
DMNet-R50 [18]	42	2736	28.2	47.6	28.9	19.9	39.6	55.8	42M	137	-
DMNet-R101 [18]	42	2736	28.5	48.1	29.4	20.0	39.7	**57.1**	60M	186	-
CDMNet-R50 [19]	12	2718	29.2	49.5	29.8	20.8	40.7	41.6	42M	137	-
CDMNet-R101 [19]	12	2718	29.7	50.0	30.9	21.2	41.8	42.9	60M	186	-
CZ-Det-R50 [37]	12	1187	33.2	58.3	33.2	26.1	42.6	43.4	-	-	-
NE-CDMNet-R50	72	4658	35.3	59.0	35.3	27.1	46.1	51.9	43M	69	2.31
NE-CDMNet-R101	72	4658	**35.9**	**59.4**	**35.9**	**27.3**	**46.6**	53.1	76M	86	1.78

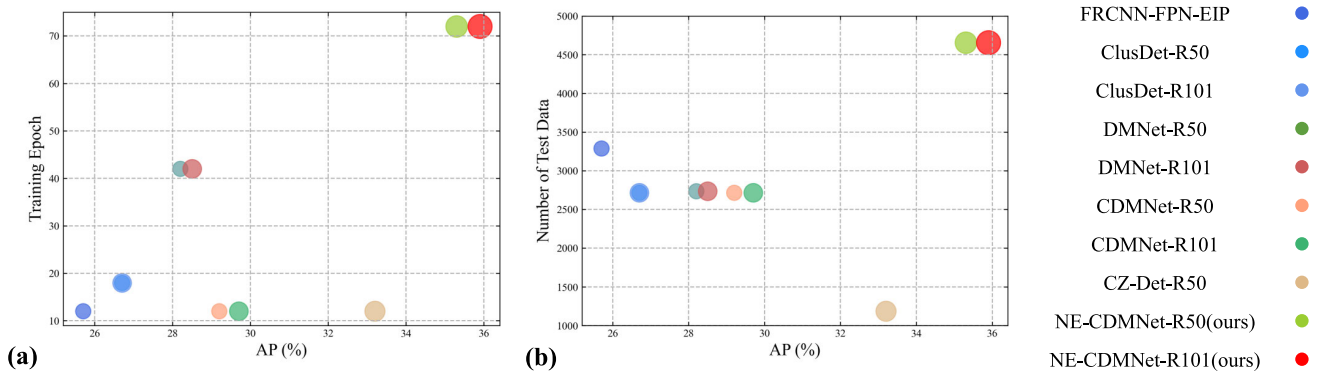


Fig. 6. Performance of different methods on VisDrone-DET2019 dataset. (a) Relationship between training epoch and AP. (b) Relationship between the number of test data cropping and AP.

(SGD) is employed as the loss optimization function in the experiments, with the momentum set to 0.9, the weight decay set to 0.0001, the initial learning rate set to 0.02, and the multistep method used to adjust the learning rate. In the training process, the foreground–background class imbalance problem in the object detector is solved using the focus loss [8], and a multiscale image input strategy is utilized. On the VisDrone-DET2019 [3] dataset, the training epochs is set to 72, and the batch size is 2. On the DIOR [4] dataset, we trained a model for 72 epochs with a batch size of 4. We used Deformable DETR as the basic detector. We report results with two different backbones: ResNet-50 [42] and ResNet-101 [42] pretrained on ImageNet-1k [12].

C. Evaluation Metrics

We follow the evaluation protocol on the MS-COCO [10] dataset and use the average precision (AP) under different intersection over union (IoU) thresholds to evaluate the detection results of our method. Specifically, AP_{50} means that the IoU threshold defined as a true positive (TP) sample is 0.5, and AP_{75} means that the IoU threshold defined as a TP sample is 0.75. AP represents the average value from AP_{50} to AP_{95}, with the IoU interval being 0.05. Furthermore, we use AP_S, AP_L, and AP_L to evaluate the detection performance of small objects, medium objects, and large objects, respectively. The relevant formulas are listed as follows:

$$\mathrm{AP} = \frac{1}{n}\sum_{k=1}^{n} J(P, R)_k \tag{13}$$

where n represents the number of categories, $J(P, R)_k$ represents the AP function, P represents the precision, and R represents the recall. The precision P and recall R defined as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{14}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{15}$$

where TP represents the true positive sample, FP represents the false positive sample, and FN represents the false negative sample.

D. Comparisons With the State of the Art

The performance of the proposed method and other state-of-the-art detectors on VisDrone-DET2019 [3] and DIOR [4] is comprehensively compared in this section.

1) *Results on the VisDrone-DET2019 Dataset:* We compare with existing uniformly cropped and other cropped method state-of-the-art detectors with different backbones, including Faster R-CNN (FRCNN) [7] with evenly image partition (EIP), ClusDet [17], DMNet [18], CDMNet [19], and cascaded zoom-in detector (CZ-Det) [37]. Results are shown in Table I and Fig. 6. The AP of our method is improved by 10.2%, 9.2%, 9.2%, 7.7%, 7.4%, 6.7%, 6.2%, and 2.7%, respectively. These methods mainly achieve selective image cropping through additional learnable modules to improve detection performance. In the end-to-end

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Fig. 7. Visualizations of predictions using the proposed method on the VisDrone-DET2019 dataset. The first row is a normal exposure image, the second row is an under-exposed image, and the third row is an over-exposed image.

TABLE II

COMPARISON WITH STATE-OF-THE-ART METHODS ON DIOR DATASET. THE BEST RESULTS ARE MARKED BOLD

Model	Backbone	APL	APO	BF	BC	BR	CH	DAM	ESA	ETS	GC	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP
FRCNN [7]	ResNet-50	62.8	26.8	71.7	80.9	34.2	72.6	18.9	65.8	66.4	66.6	79.2	34.9	48.8	81.1	64.3	71.2	87.4	47.3	50.5	65.2	56.6
LO-Det [43]	MobileNetV2	72.6	65.0	76.7	84.7	33.7	73.7	56.8	75.9	57.5	66.3	68.0	60.9	51.5	88.6	68.0	64.3	86.3	47.6	42.4	76.7	65.9
Rfpoint [44]	Swin-t	77.7	59.5	79.3	89.4	45.5	80.9	40.9	84.1	76.0	83.7	82.0	43.9	60.6	88.6	70.7	78.6	86.7	66.5	53.9	69.2	70.9
YOLOv5 [22]	CSPDarknet53	81.5	78.6	78.5	88.4	46.3	73.9	59.4	83.6	69.2	78.8	82.2	60.9	57.7	74.0	78.1	71.7	82.6	53.5	56.4	80.8	71.8
RSANet [45]	DLA-34	73.6	86.0	72.6	89.6	43.6	75.3	62.3	79.5	68.7	78.6	79.1	57.9	59.2	**90.0**	55.8	77.0	87.8	65.3	55.3	86.5	72.2
AFDet [46]	DLA-34	82.4	81.5	81.9	**89.8**	51.7	74.9	58.7	**84.2**	73.3	79.5	81.0	44.2	62.0	77.8	63.2	76.9	91.0	62.5	59.3	87.1	73.2
CANet [47]	ResNet-101	70.3	82.4	72.0	87.8	**55.7**	79.9	67.7	83.5	**77.2**	77.3	83.6	56.0	63.6	81.0	79.8	70.8	88.2	67.6	51.2	**89.6**	74.3
DCN [48]	ResNet-50	90.1	76.1	92.8	83.1	45.1	90.3	63.1	67.0	60.3	75.9	81.0	56.2	65.0	79.5	87.9	81.0	88.8	62.4	72.5	77.8	74.8
YOLOX [23]	CSPDarknet53	90.0	72.0	91.9	77.9	43.4	87.8	65.3	66.5	61.3	74.0	74.8	**65.2**	62.3	89.6	90.0	79.4	88.6	54.6	70.1	70.4	74.8
DGRN [49]	Cascade RCNN	68.1	79.6	75.1	88.1	**55.7**	87.4	65.8	83.7	74.0	82.2	**92.6**	63.2	68.0	76.3	82.6	68.2	88.9	65.1	60.2	86.1	75.4
SDPNet [50]	CSPDarknet53	86.8	81.4	83.4	90.5	52.7	78.1	63.2	87.8	73.6	81.9	85.6	**65.2**	64.7	80.4	82.6	77.8	87.5	56.1	59.9	84.7	76.2
NE-CDMNet	ResNet-50	**90.8**	**90.2**	**94.1**	85.5	50.5	**90.6**	**74.3**	80.3	73.7	**84.3**	85.0	48.1	**68.2**	82.4	**94.1**	**83.8**	**91.9**	**69.3**	**74.6**	81.4	**79.7**

case, our proposed method achieves good overall performance, especially on small object detection, where we obtain the best AP, significantly outperforming all existing methods. Only on large object detection does DMNet perform better than our method. However, we do not achieve any advantage in terms of Params, GFLOPS, and FPS because our method sacrificed computing speed for end-to-end detection.

We visualize the detection results of the ground truth, the baseline model and the proposed method in Fig. 7. From the comparison results, this method can better detect small and medium-sized objects in aerial scenes.

2) *Results on the DIOR Dataset:* We have tested the proposed method on the DIOR dataset and compared it with some recent mainstream methods. The detailed results are listed in Table II. The results show that the proposed method is effective in almost all categories, reaching a comparable level as the recent state-of-the-art methods [7], [22], [23], [43], [44], [45], [46], [47], [48], [49], [50]. Experiments in Table II show that the proposed method substantially outperforms the compared methods on a subset of the categories, achieving a high mean AP (mAP). The visualizations of the proposed method are shown in Fig. 8. On categories such as APL, APO, BF, CH, DAM, GC, OP, STA, STO, TC, TS, and VE, the proposed method demonstrates an increase of at least 0.2% in performance, with the highest increase reaching almost 6.6% compared with other state-of-the-art detectors. The proposed method also achieves competitive performance in the remaining categories. Considering the overall performance, the mAP of the proposed method reaches 79.7%.

E. Ablation Study

To evaluate the contributions of different components, we conduct detailed ablation experiments on two benchmark datasets. The results are shown in Table III. The table shows that we compare four variants: 1) using baseline for performance evaluation; 2) introducing the DMC model and LGF algorithm based on variant I; 3) using the new QS scheme in the baseline; and 4) performance evaluation using our method.

1) *Effectiveness of DMC:* Section I discussed the pros and cons of uniform cropping, random cropping, and selective cropping methods. The proposed method designs an end-to-end selective cropping method based

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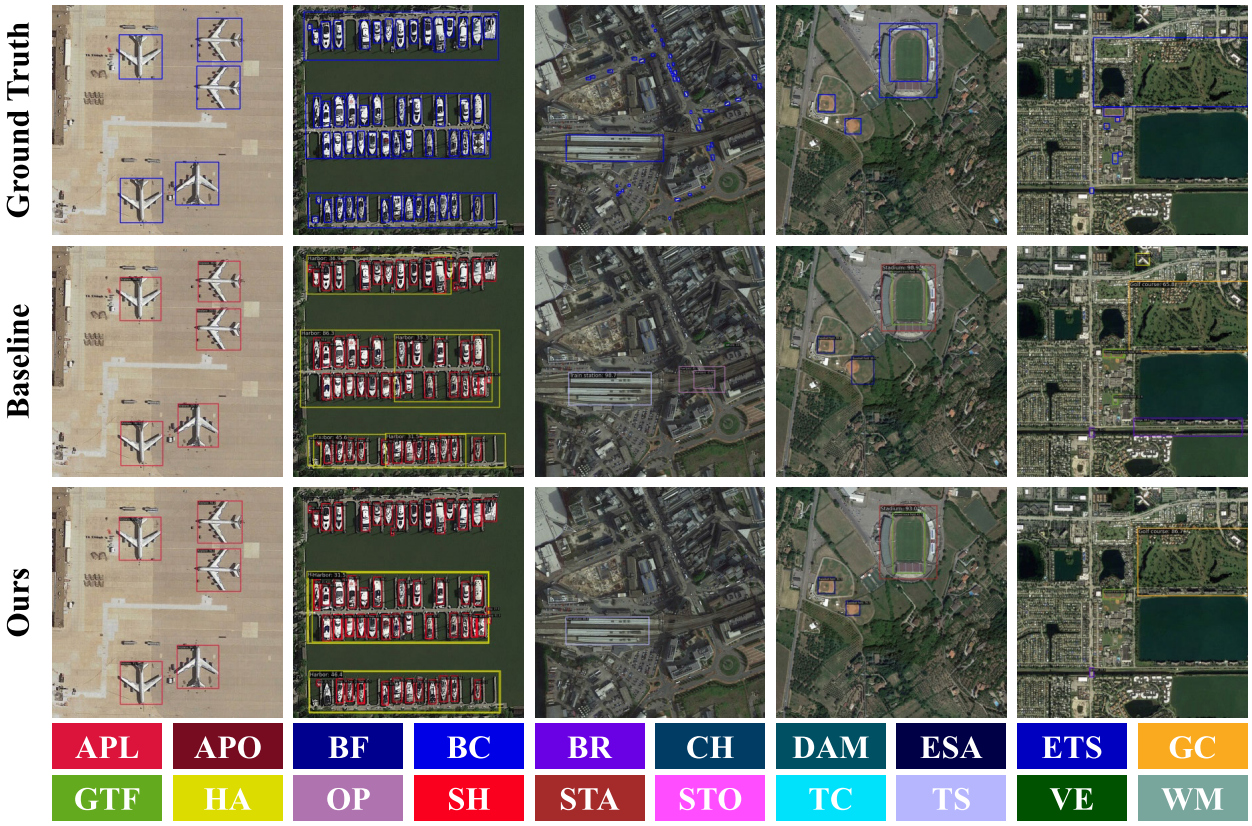


Fig. 8. Visualizations of predictions using the proposed method on the DIOR dataset.

TABLE III

ABLATION STUDY RESULTS ON THE VISDRONE-DET2019 DATASET AND THE DIOR DATASET. THE NUMBERS IN () INDICATE IMPROVEMENTS RELATIVE TO THE BASELINE

	DMC	QS	LGF	AP on VisDrone-DET2019(%)	mAP on DIOR(%)	FLOPS(G)	Params(M)
I				31.0	75.8	69	42
II	√		√	33.3(+2.3)	78.0(+2.2)	69	43
III		√		33.7(+2.7)	77.3(+1.5)	69	42
IV	√	√	√	35.3(+4.3)	79.7(+3.9)	69	43

TABLE IV

RESULTS OF USING DIFFERENT CROPPING METHODS. THE BASELINE MODEL IS FRCNN

Methods	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L
Baseline [7]	21.4	40.7	19.9	11.7	33.9	54.7
Uniform Cropping	21.1	44.0	18.1	14.4	30.9	30.0
Cluster [17]	26.7	50.6	24.7	17.6	38.9	51.4
DMNet [18]	25.9	-	-	19.4	38.1	41.6
Ours	27.3	51.4	25.1	20.1	37.9	55.1

on the selective cropping method and realizes the end-to-end definition by introducing the DMC method without additional components in Section III-B. Table IV shows the results of using different cropping methods under the same baseline. We notice that while the overall performance in terms of AP improves after applying uniform cropping and other cropping strategies, the performance on large objects (AP_L) actually decreases. This is because the uniform and other cropping methods truncate large objects into fragmented regions, resulting in many false positive detections. The uniform and other cropping techniques do not differentiate between important and less important regions in the input. As a result, they can inadvertently crop and partition large, salient objects, causing the model to detect these objects as multiple smaller fragments. This fragmentation leads to an increase in false positive predictions, consequently degrading the detection performance on the large object categories. In contrast, the proposed selective cropping method aims to address this limitation. By identifying and preserving the important regions based on the density map, the selective cropping approach is able to maintain the integrity of large objects during the cropping process. This helps to retain the model's ability to accurately detect large objects, while still benefiting from the overall performance improvements enabled by the selective cropping strategy. Specifically, after introducing our selective cropping method, the performance of AP is increased by 5.9%, while AP_S, AP_M, and AP_L are all positive gains.

2) *Effectiveness of QS Scheme:* Compare with the baseline method in Section III-C, we have designed a new QS

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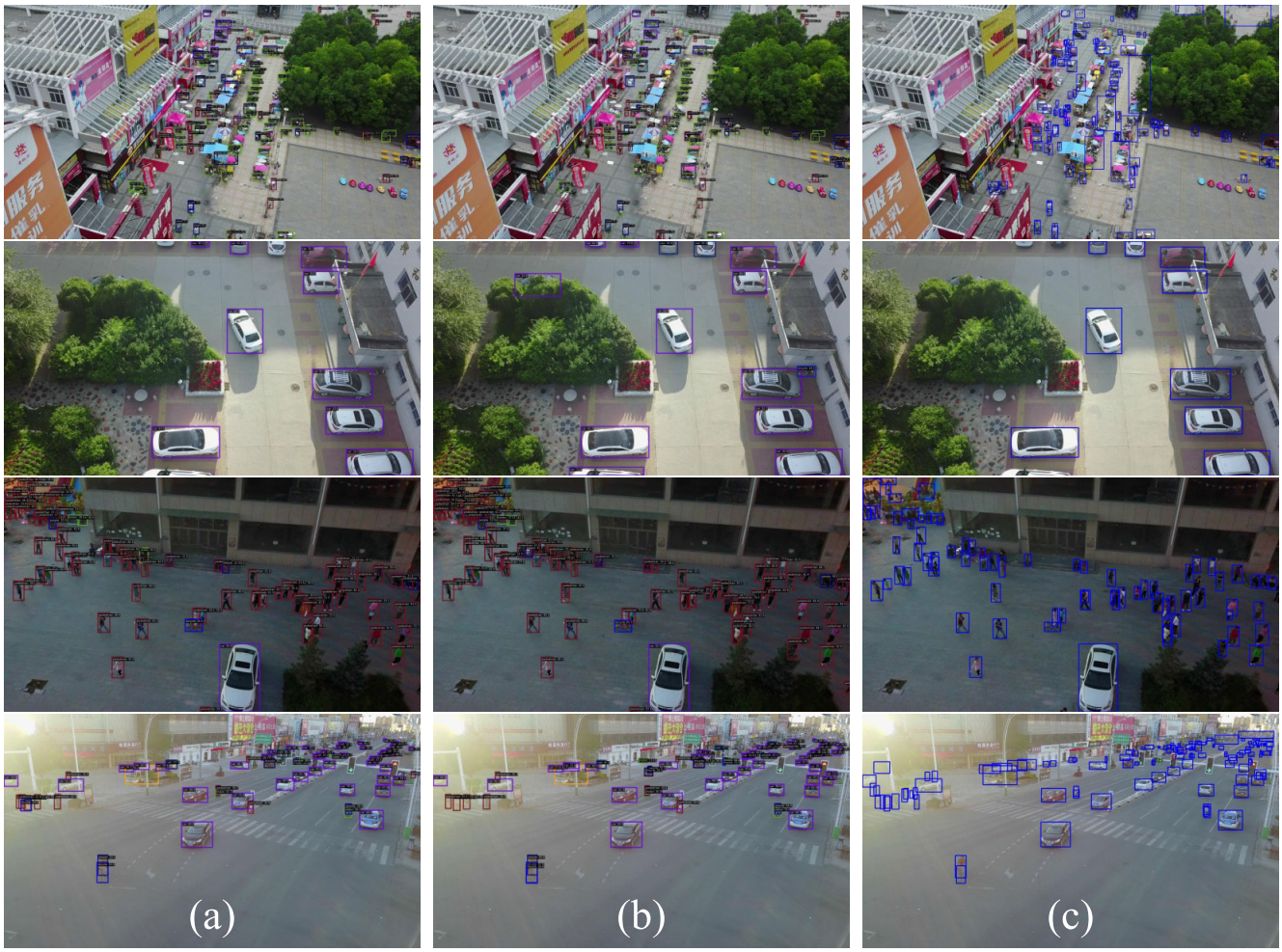


Fig. 9. Visualizations of the effect of using the DMC method with and without the fusion algorithm. (a) Without the fusion algorithm, (b) with the fusion algorithm, and (c) ground-truth image.

scheme to address the difficulty of query optimization. The baseline method directly selects features from the encoder, which may be ambiguous and misleading for the decoder. For example, a selected feature may contain multiple objects or be only part of an object. In contrast, our proposed method uses confidence scores to select the top-K features from the encoder to initialize the position query, making the content query as learnable as before, to help the model better utilize the position information to obtain more comprehensive content features from the encoder. Overall, by combining the learned content query with the selected top-K position query, our method enables the model to better leverage the spatial and semantic information from the encoder to generate more accurate and comprehensive outputs. Our QS scheme addresses the limitations of the baseline method, thereby improving the performance. As shown in Table III, the QS scheme outperforms the baseline 2.7%AP (33.7 versus 31.0) and 1.5%mAP (77.3 versus 75.8), which demonstrates the effectiveness of this scheme.

3) *Effectiveness of LGF Algorithm:* The LGF algorithm proposed in NE-CDMNet aims to combine the detection results from the original image and the density cropping image. In Section III-D, we discussed the importance of this LGF algorithm. After obtaining the initial detection results from the model, the next step is to fuse the outputs from the original image and the density cropping image. This means that the LGF algorithm and the DMC model need to be used in tandem. As shown in Table III, we report the accuracy of the AP metrics under a fusion threshold of 0.55. This joint application of the LGF algorithm and the DMC allows the model to effectively combine the local and global results, leading to more accurate final detection results. Furthermore, Fig. 9 visualizes the effect of using the DMC method with and without the fusion algorithm. It can be clearly seen that the detection performance is significantly worse without the fusion algorithm. The fusion step is crucial to integrate the bounding box position predictions in the original image and the cropped image, which can improve the overall detection quality.

F. Discussion

1) *Confusion Matrix on DIOR:* The DIOR dataset is one of the largest, most diverse, and publicly available object detection datasets in the Earth observation community.

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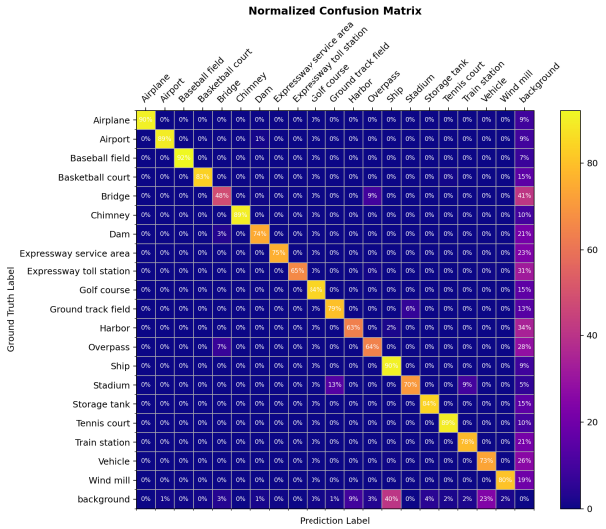


Fig. 10. Confusion matrix of our method on the DIOR dataset. The rows of the confusion matrix indicate the GT labels, the columns indicate the predicted categories, the last column indicates the missed detection rate for each category, the bottom row indicates the false detection rate for different categories and the values on the diagonal of the confusion matrix represent the accuracy of predictions for each category.



Fig. 11. Visualizations of the primary confusion errors occur between two similar categories (BR and OP, as well as STA and GTF).

A key characteristic of this dataset is the significant variation in instance sizes. To further analyze the performance of our NE-CDMNet method, we present the confusion matrix in Fig. 10. As illustrated in the results, Our NE-CDMNet method demonstrates high accuracy for most object categories. The missed detection rate and false detection rate are also relatively low, with the false detection rate of most categories below 10% and the missed detection rate below 25%. The primary confusion errors occur between two similar categories (BR and OP, as well as STA and GTF). In these cases, the model sometimes misidentifies one category as the other, as shown in Fig. 11. In addition, there are a few instances where the foreground object is not recognized, or it is recognized as background. Through this detailed confusion matrix analysis, we can gain valuable insights into the strengths and weaknesses of our NE-CDMNet method on various categories in the DIOR dataset. It highlights the categories where our method excels, as well as the areas where further improvements can be made. This information is extremely valuable in guiding future research (coarse–fine grained detection and attention mechanism) and optimization of our object detection methods. It allows us to focus our efforts on addressing the specific challenges and confusions identified, ultimately leading to more robust and accurate object detection performance across the full spectrum of object categories in the DIOR dataset.

2) *Limitation of NE-CDMNet:* Although we proposed an end-to-end selective cropping detection method that breaks the original situation where additional components are involved in training, there are still some shortcomings. As shown in Tables I and II, the experimental results verify the superiority of our method in terms of accuracy and adaptability. Although end-to-end aerial image detection is achieved, the model parameter volume reaches 76 M, which is still not convenient for deployment on embedded devices. In subsequent research, we will explore lightweight end-to-end aerial images object detection methods to reduce model size and improve its applicability in resource-constrained environments. Specifically, by transferring knowledge from the teacher model to the student model to improve performance; using pruning to remove the smallest contributing part of the parameters in each layer of the network; using quantization to truncate long floating-point numbers to integers expressed in low bits.

V. Conclusion

In this article, we propose an end-to-end selective cropping detection method that breaks the original situation where additional components are involved in training. First, we use a no-extra components DMC method to avoid occluding or truncating the object, aiming to solve the problem that other cropping methods cannot distinguish between important and unimportant areas. Second, we design a new QS scheme to help the model better utilize the position information to obtain more comprehensive content features from the encoder. Finally, we apply the LGF algorithm to all bounding boxes and calculate the final result. Extensive experimental results on two widely used public datasets in aerial image object detection (VisDrone-DET2019 dataset and DIOR dataset) demonstrate the effectiveness of our method. Although no-extra components DMC guided object detection is achieved, it is still not convenient to deploy on embedded devices. In the future, we will further explore lightweight end-to-end aerial

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image object detection methods and extend them to the field of real-time detection.

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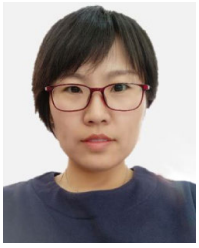
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